UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. What does the % operator return?
   A. The absolute value
   B. The remainder of integer division
   C. The quotient
   D. The square root

2. What is displayed by the following code?

```
int x = 5;
cout << x * 2;
```

   A. 10
   B. 5
   C. 25
   D. 7

3. Which identifier is valid in C++?
   A. total score
   B. total_score
   C. 2total
   D. int

4. Given the declaration, what is displayed?

```
char word[6] = {'p', 'r', 'i', 'n', 't', '\0'};
cout << word[3];
```

   A. p
   B. n
   C. t
   D. i

5. Which of the following are selection structures?
   i. if
   ii. switch
   iii. for
   iv. while
   A. iii and iv
   B. i and ii
   C. ii and iv
   D. i and iii

6. Which statement immediately exits a loop?
   A. continue
   B. return 0
   C. cin
   D. break

7. Which statement is most suitable for selecting among several constant cases?
   A. do...while
   B. switch

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C. while
D. for

8. What does endl do?
A. Declares an integer
B. Writes a newline and flushes the stream
C. Ends the program
D. Reads a line

9. Which of the following declarations are valid?
i. int a[5];
ii. int b;
iii. int c[3] = {1, 2, 3};
iv. int d[];
A. iii and iv
B. i and ii
C. i and iii
D. ii and iv

10. What is pseudocode?
A. A type of compiler error
B. A C++ header file
C. Machine code produced by an assembler
D. A structured English-like description of an algorithm

PART B (60 MARKS)

QUESTION 1

This question tests high-level languages and debugging.

a) Define syntax error and logic error.

(2 marks)

b) Explain TWO advantages of high-level languages over machine language.

(3 marks)

QUESTION 2

A program reads a student's score from 0 to 100 and displays the grade shown in Table 1.

Table 1. Grade Scale

Score	Grade
80-100	A
60-79	B
50-59	C
40-49	D
0-39	F

a) Draw a flowchart for the grade classification.

(6 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

(7 marks)

QUESTION 3

This question tests type conversion and constants.

a) Explain the difference between implicit and explicit type conversion, and give one C++ example of each.

(5 marks)

b) Write a C++ program segment that calculates the average of three integer marks as a double and displays it to two decimal places.

(6 marks)

QUESTION 4

Study the following nested loop:

```
for (int i = 1; i <= 3; i++) {
    for (int j = 1; j <= i; j++) {
        cout << j;
    }
    cout << endl;
}
```

a) Trace the program and show the exact output.

(10 marks)

b) Identify TWO logic errors in a version that prints four rows but reuses j without resetting it. Rewrite the corrected code.

(5 marks)

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QUESTION 5

A sports club records the sport names and medal counts in Table 1.

a) Declare two parallel arrays for the five sports in Table 1 and initialise them with those values.

Table 2. Sport and Medal Records

Sport	Medals
Swimming	8
Badminton	12
Athletics	7
Cycling	5
Hockey	9

(8 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

(8 marks)

PART C (30 MARKS)

QUESTION 1

A catering company charges the meal prices shown in Table 1.

Table 3. Meal Price

Quantity Range	Price per Meal (RM)
50 - 99	11.50
100 - 199	10.50
200 and above	9.50

Write a complete C++ program that reads the customer name and quantity, validates the quantity, calculates the total, displays a receipt, and repeats until the user chooses to stop. At the end, display the total quantity and total sales.

(15 marks)

QUESTION 2

A bookstore records up to five books. Table 1 shows an example of the fields.

Table 4. Book Record Fields

Title	Price (RM)	Quantity
Book A	45.00	3
Book B	30.00	5
Book C	55.00	2

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

END OF QUESTION PAPER

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